

Short-Run cost structure of firms

Introduction:

The cost of production is very important factor while determining the quantity of goods that will be produced and supplied. If the production cost is lower than the price or equal to it, then the firm will supply more. Those firms which are not capable to cover their cost of production will exit the market sooner or later.

Cost functions are derived from the production functions. It describes the relationship between the total cost and the quantity of output, assuming other determinants of cost to be constant. Other determinants of cost may comprises technology, prices of factors, size and utilization of plants, lot size, efficiency, stability of output etc. When the relationship between cost and output is shown graphically, it becomes cost curve. If there is any change in the determinants of costs, then the cost curve shifts. In this module, short run cost curves is explained. Short run is a time period in which firm can vary its variable inputs only.

Cost in the Short Run:

Total Fixed Cost (TFC) In the short run, some inputs are fixed, while others are variable. Cost incurred by the firm on the employment of fixed inputs in the short period is called fixed cost. This cost must be incurred even if the level of output is zero. Thus, fixed cost are those costs which do not changes with the changes in the level of output. For example: salaries of administrative staff, rent, property taxes, depreciation of machinery, interest on bonds etc. This cost is also called as overhead cost or indirect cost or supplementary cost.

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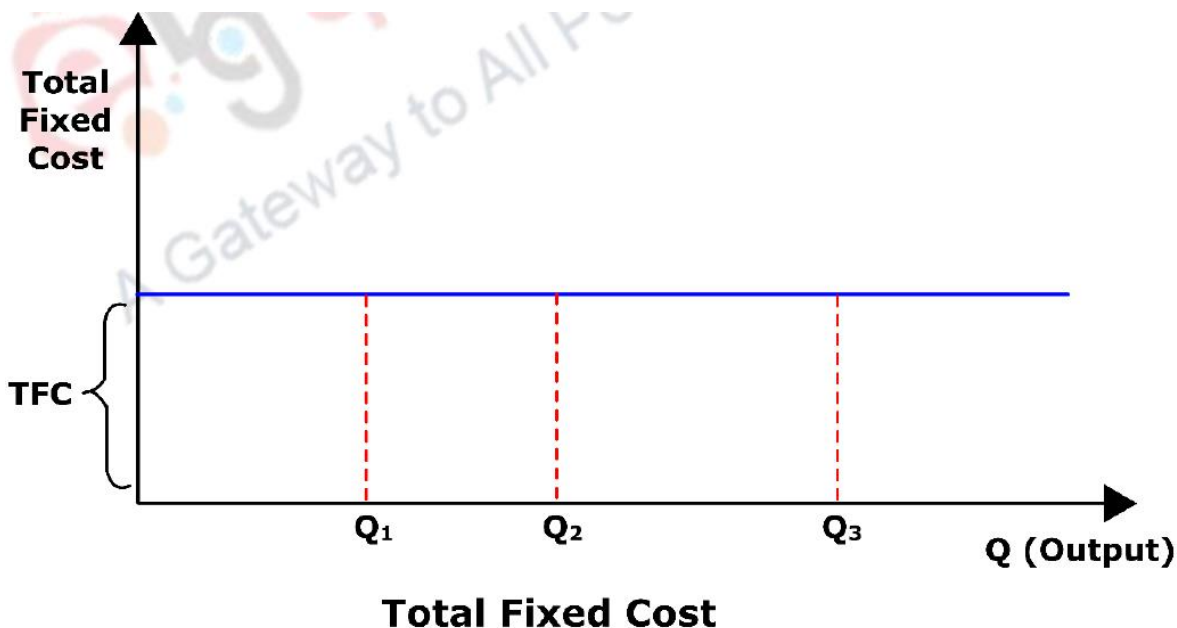
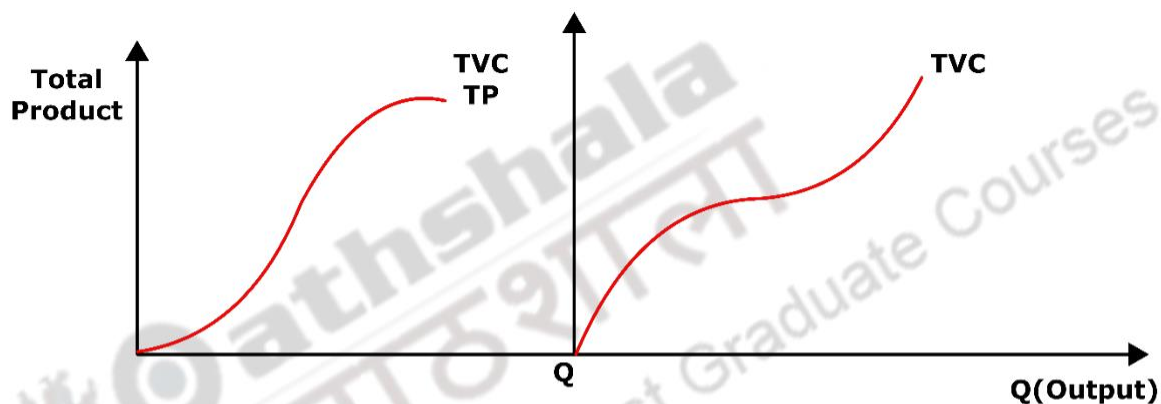


Figure 1.1:

In fig 1.1, Total fixed cost is a horizontal line parallel to X axis. This shows that total fixed cost remains constant irrespective of the level of output.

Total Variable Cost (TVC)

Variable cost is the cost of employing variable inputs in production like raw material, labour, fuel, electricity, transportation, sales commission etc. The variable cost varies directly with the change in the level of output. If the firm wants to increase the level of production in the short run, it can do so by employing additional variable factors. Therefore, variable cost will rise, when output increases and vice-versa. Total variable cost is zero when the level of output is zero and rise with an increase in output but the rate of increase is not same. TVC is shown in figure 1.2.



Total Variable Cost Curve

Figure 1.2

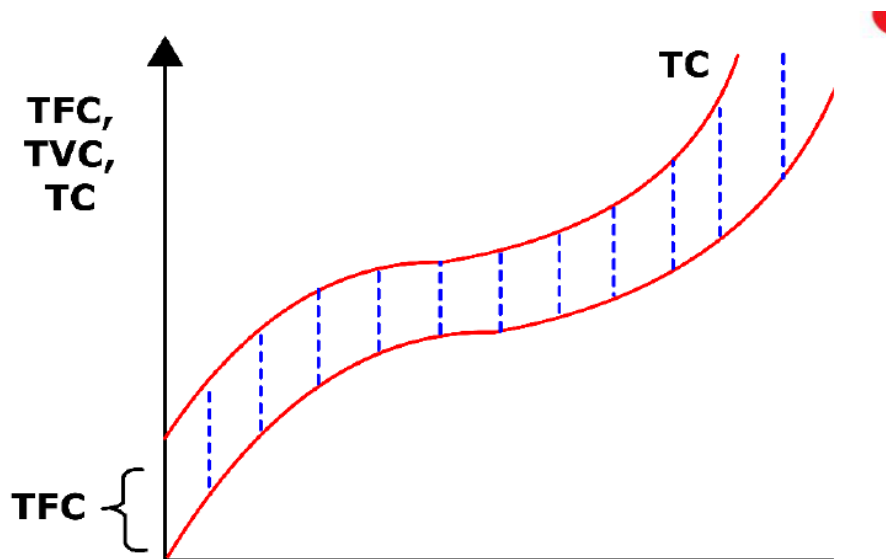
TVC is an inverse S-Shaped curve reflecting the operation of law of variable proportions. According to the law, when the firm increases the quantity of variable factor, keeping all other factors as fixed then total product increases at an increasing rate implying TVC increases at a declining rate. After point A, total product increase at a decreasing rate which implies that TVC increasing at an increasing rate.

Total Cost (TC)

Total cost is the aggregate of total fixed cost and total variable cost.

$$TC = TFC + TVC$$

Total cost is the cost incurred on the employment of fixed and variable factors to produce any given level of output. If there is a change in the ratio of output to input, then total cost will also change. TC is shown in figure 1.3.



Total Cost Curve

Figure 1.3:

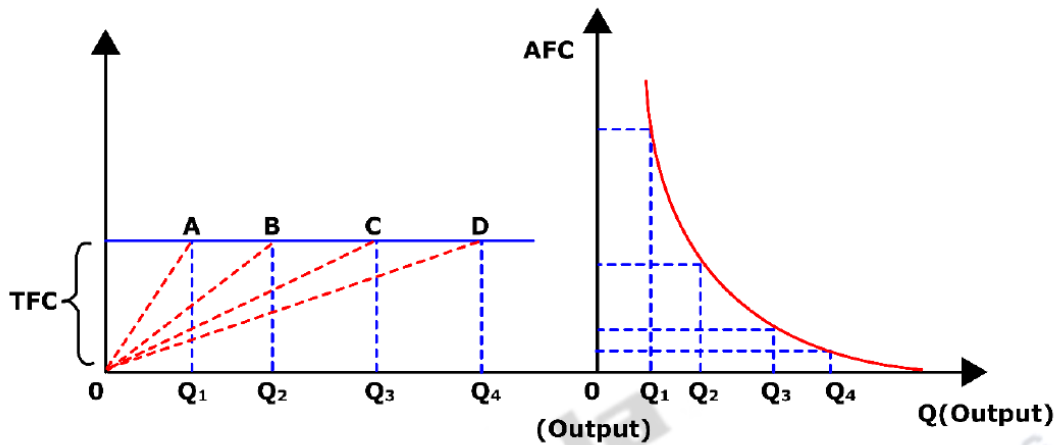
In fig. 1.3, TC is broadly an inverse S- Shaped curve. It is obtained by summing vertically TFC and TVC curves. Total cost rises with an increase in the level of output, as it depends more on total variable cost. The vertical distance between total cost curve and total variable cost curve is same throughout representing TFC. The vertical distance between TC and TFC is TVC which rises with the increase in the level of output.

Average Fixed Cost (AFC)

Average fixed cost is the per unit fixed cost of producing a product. It is calculated by dividing TFC by the quantity of output produced.

$$AFC = \frac{TFC}{X}$$

Total fixed costs are constant. There is a negative relationship between AFC and output. As the level of output increases, AFC curve falls continuously but it never touches either of the axes. Graphically, AFC curve is derived from the TFC curve which is shown in fig 1.4.



Derivation of AVC curve from the TFC Curve

Figure 1.4

The AFC at a particular point is given by the slope of the ray from the origin that touches the TFC line at given level of output on the x-axis. In fig. 1.4, AFC at point 'A' on the TFC curve is the slope of the ray OA. Similarly, we can obtain other points like 'B' and 'C' and 'D'.

AFC curve is a rectangular hyperbola which shows that as we increase the level of output, fixed cost per unit decreases:

Average Variable Cost (AVC)

Average variable cost is the per unit variable cost of producing a product. It is calculated by dividing TVC by the quantity of output produced.

$$AVC = \frac{TVC}{X}$$

The average variable cost falls initially as the level of output increases due to increasing returns. It then reaches a minimum point at optimal level of output. Beyond this minimum point, AVC rises steeply because of diminishing returns.

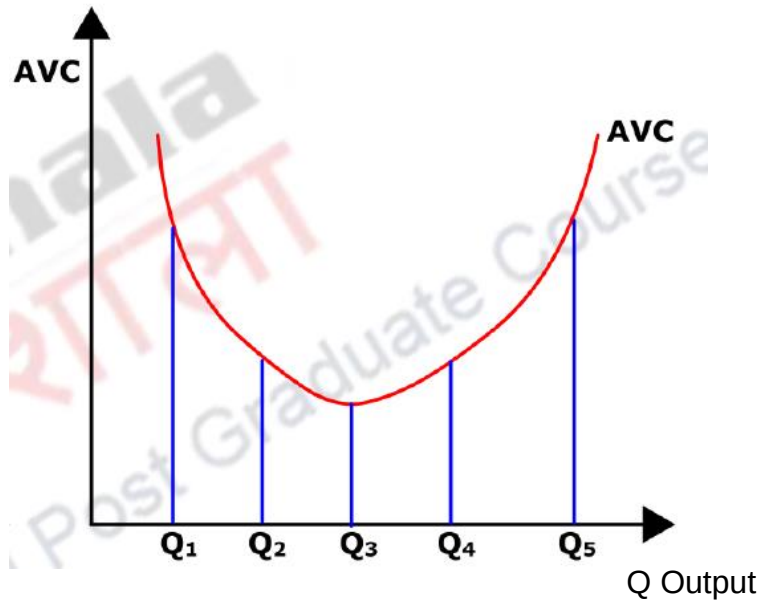


Figure 1.5: AVC curve

Average Cost (AC)

Average cost is the summation of average fixed cost and average variable cost. It is calculated by dividing TC by the quantity of output produced.

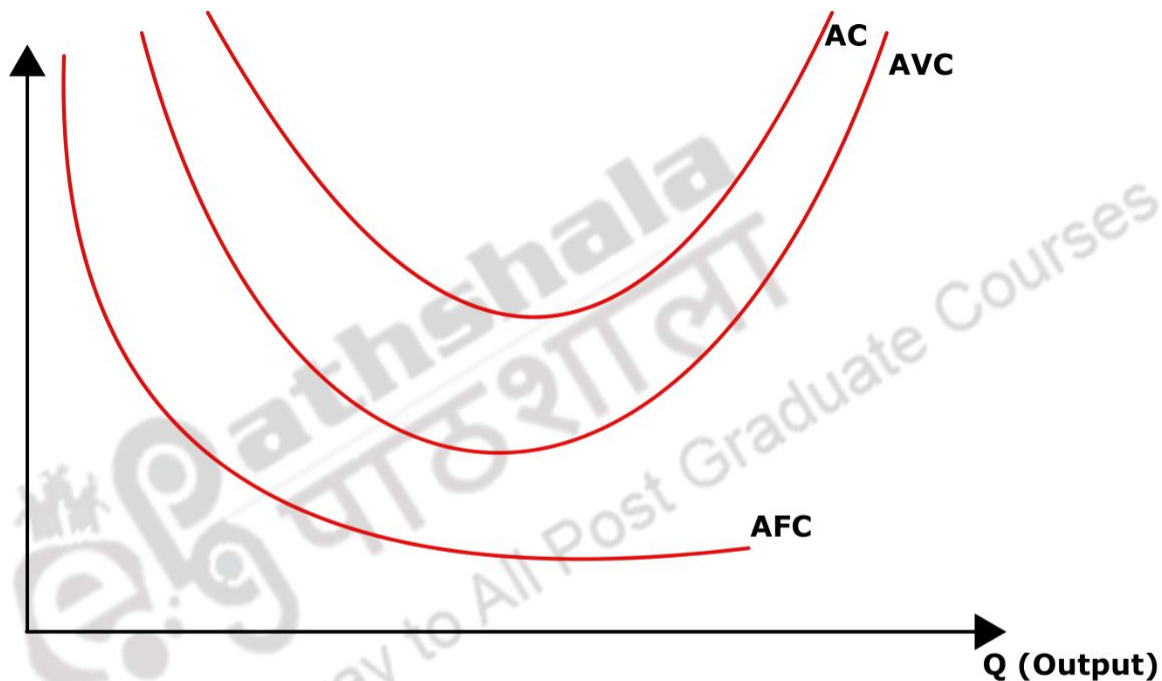
We know that :

$$TC = TFC + TVC$$

Divided by Q then, $AC = AFC + AVC$

$$AC = \frac{TC}{Q} ; AVC = \frac{TVC}{Q} ; \text{ and } AFC = \frac{TFC}{Q}$$

Average cost is shown in figure 1.6.



Derivation of AC curve from the TC Curve

Figure 1.6: Derivation of AC curve from the TC Curve

Initially average total cost falls sharply with the increase in the level of output as both AVC and AFC curves fall. It then reaches a minimum at the optimum level of output and then AC rises.

Marginal Cost (MC)

Marginal Cost is the change in total cost or total variable cost when the output is increases by one unit. For example: the total cost of producing 7 units of a product is Rs. 100 and that of 8 units is Rs. 120, then marginal cost of producing 8th unit is Rs.120- Rs.100= Rs.20.

$$MC = \frac{d(TC)}{dX}$$

$$MC = \frac{d(TVC)}{dX}$$

MC curve is U shaped curve. Initially when the level of output increases, MC falls continuously due to efficient use of variable factor and thus rises sharply.

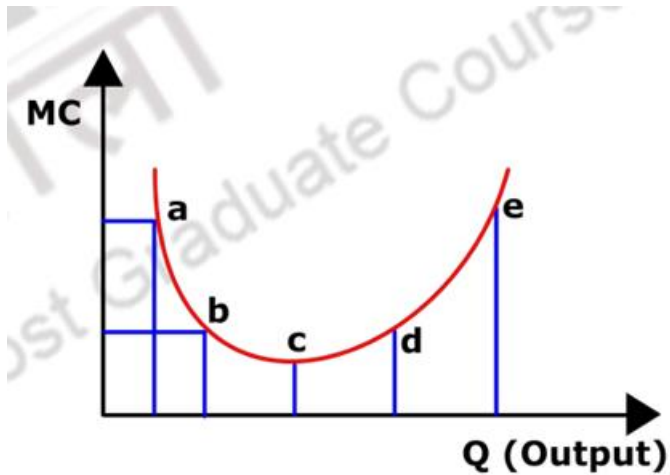
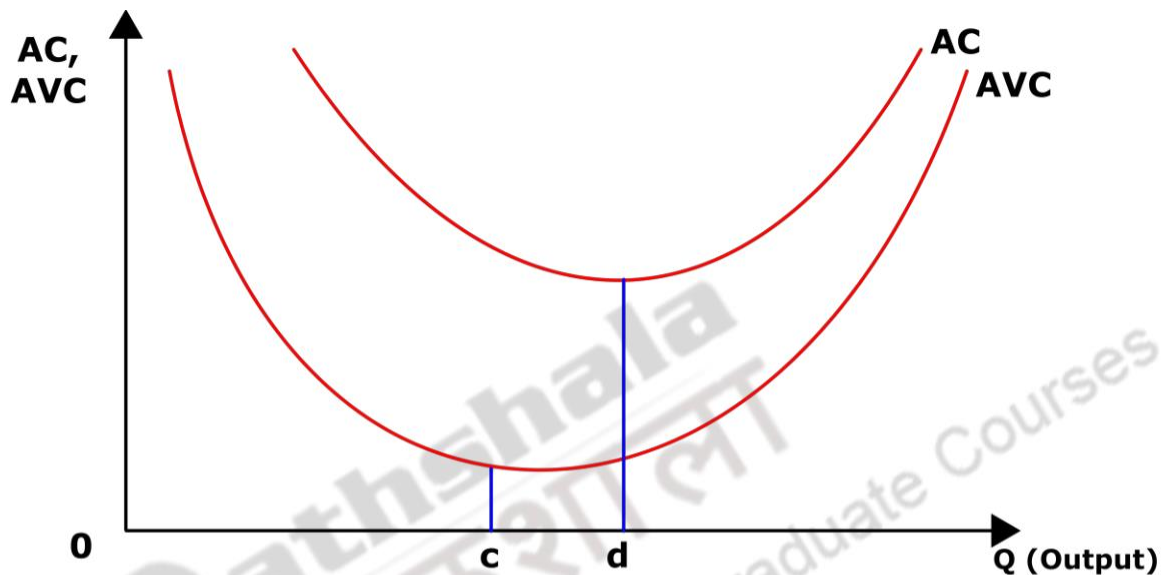


Figure 1.7: MC curve

Relationship Between AC and AVC:



Relationship between AC and AVC

Figure 1.8: Relationship between AC and AVC

Both AC and AVC are U Shaped curve because of operation of law of variable proportions. In fig 1.8, minimum point of AC (point d) takes place to the right of minimum point of AVC i.e. point c. This is due to the reason that AC comprises AVC and AFC curves and both are falling initially with the rise in level of output.

After the AVC attains its minimum point at b, corresponding to the quantity OC1 , the AVC begins rising. Between c and d , the rate of fall in AFC offsets the rate of rising AVC resulting continuous fall in AC. At Xc level of output, the rate of rise of AVC is equals to the rate of fall of AFC resulting AC to reaches its minimum point. Beyond Xc level of output, the rate of rise in AVC is greater than the rate of falling AFC, causing the AC to starts rising.

With the increase in the level of output, AVC approaches AC asymptotically. This is because AFC continuously falls as output increases.

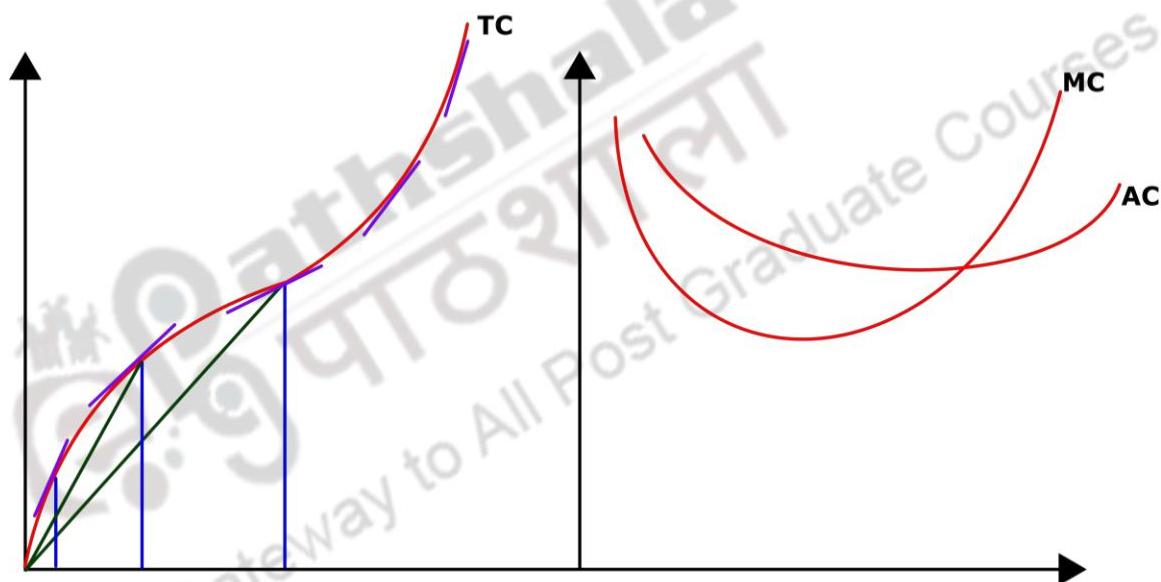
Relationship Between AC and MC:

Both AC and MC curves are U shaped curves and are derived from TC curve. Algebraically,

$$AC = \frac{TC}{X}$$

$$MC = \frac{d(TC)}{dX}$$

MC at each point on the TC curve is the slope of the TC curve at that point and AC is the slope of the ray starting from the origin to that point on TC curve. The unique relationship between AC and MC curve is shown in Figure 1.9.



Relationship between AC and MC curves

Figure 1.9: Relationship between AC and MC curves

1. When AC is falling with the increase in the level of output, MC will be lower than AC and lies below it.
2. When AC is rising with the increase in the level of output, MC will be higher than AC and lies above AC.
3. At the optimum level of output, AC is minimum and constant (i.e. neither rising nor falling). MC is equal to AC.
4. MC curve cuts the AC curve at its minimum point.

Summary:

- The fixed cost are those costs which do not change with the changes in the level of output. For example: salaries of administrative staff, rent, property taxes, depreciation of machinery, interest on bonds etc.
- The variable cost varies directly with the change in the level of output.
- TVC is an inverse S-Shaped curve reflecting the operation of law of variable proportions.
- Total cost is the aggregate of total fixed cost and total variable cost.
- Average fixed cost is the per unit fixed cost of producing a product. It is a rectangular hyperbola.
- Average variable cost is the per unit variable cost of producing a product. It is an U-Shaped curve.
- Average cost is the summation of average fixed cost and average variable cost. It is calculated by dividing TC by the quantity of output produced.
- Marginal Cost is the change in total cost or total variable cost when the output is increased by one unit.
- MC curve is U shaped curve. Marginal cost at any quantity of output is derived by the slope of the TC or TVC curve.
- With the increase in the level of output, AVC approaches AC asymptotically. This is because AFC continuously falls as output increases.